



Automatic Extraction of Electoral Results from the Tally Sheets of the 2026 Peruvian General Elections

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Abstract—Official electoral results in Peru are published as scanned tally sheets (*actas*), which prevents large-scale automatic processing. We present an end-to-end pipeline that downloads *actas* from the official ONPE portal, locates 46 fields per document through a fixed template anchored by fiducial-mark registration, and reads handwritten vote counts with digit-restricted OCR. Evaluation is annotation-free: ground truth comes from the officially digitized results published by the same portal, and the test set is a seeded random national sample (100 *actas*, 23 departments). The population is heterogeneous: 24% are born-digital documents whose text layer yields 95.1% field accuracy without vision. On 76 scanned handwritten *actas* (3,268 fields), the pipeline attains 59.6% field accuracy (95% CI [56.6, 62.5]) and 0.401 CER, against 48.4% for an end-to-end multimodal LLM prompted few-shot, quantifying the ceiling of generic recognition and motivating fine-tuned recognizers trained on the ~ 43 field-value pairs each *acta* provides for free.

Index Terms—document analysis, optical character recognition, handwritten digit recognition, electoral documents, template registration

Resumen—Los resultados electorales oficiales del Perú se publican como actas de escrutinio escaneadas, lo que impide su procesamiento automático a gran escala. Presentamos un pipeline de extremo a extremo que descarga las actas del portal oficial de la ONPE, localiza 46 campos por documento mediante una plantilla fija anclada por registro de marcas fiduciales, y lee los conteos manuscritos con OCR restringido a dígitos. Dos decisiones de diseño hacen la evaluación reproducible y sin anotación manual: la verdad de terreno proviene de los resultados oficialmente digitados que publica el mismo portal, y el conjunto de prueba es una muestra aleatoria nacional con semilla publicada (100 actas, 23 departamentos). La población de actas es heterogénea: el 24% son documentos nacidos digitales cuya capa de texto rinde 95.1% de exactitud por campo sin visión alguna. Sobre las 76 actas manuscritas escaneadas (3,268 campos), el pipeline alcanza 59.6% de exactitud por campo (IC 95% [56.6, 62.5]) y CER de 0.401, frente a 48.4% de un LLM multimodal de extremo a extremo con few-shot, cuantificando el techo del reconocimiento genérico y motivando reconocedores afinados con los ~ 43 pares campo-valor que cada *acta* provee gratuitamente.

Palabras clave—análisis de documentos, reconocimiento óptico de caracteres, reconocimiento de dígitos manuscritos, documentos electorales, registro de plantilla.

I. INTRODUCTION

Scrutiny tally sheets (*actas de escrutinio*) are the official source of electoral results for each polling station in Peru. Although the National Office of Electoral Processes (ONPE) publishes every *acta* digitally, the information they carry is visual: a scanned form where members of the polling station wrote vote counts by hand. Manual extraction is slow, error-prone and does not scale to the $\sim 88,000$ polling stations of a general election, which limits automatic auditing and independent verification.

This work asks a concrete question: *how far can a classical computer vision pipeline go on real, nationally sampled Peruvian actas, and what does that imply for the design of a production system?* We build and evaluate a five-stage pipeline (acquisition $\rightarrow$ preprocessing $\rightarrow$ region detection $\rightarrow$ recognition $\rightarrow$ structured output) on the first round of the 2026 General Elections. Our contributions are:

- 1) **An annotation-free, reproducible evaluation protocol.** The same portal that publishes each *acta* image also publishes its officially digitized vote counts through an internal API. We use those values as ground truth, which turns every downloaded *acta* into ~ 43 labeled field-value pairs at zero annotation cost, and we sample the test set randomly at national scale with a published seed.
- 2) **Template detection made robust by fiducial registration.** The *acta* is a fixed-layout form with printed registration marks. We show that fixed fractional templates break under real-world scanner framing (black bands, offsets) and that a RANSAC-based translation matching of fiducial marks followed by a partial affine warp recovers alignment, adding +8.7 accuracy points.
- 3) **A population finding with architectural consequences.** In a random national sample, 24% of *actas* are born-digital (STAE) documents whose PDF text layer can be parsed directly with 95.1% field accuracy; only 76% are scanned handwritten forms that require vision. A production system should classify the *acta* type first and route accordingly.
- 4) **A quantified ceiling for generic OCR.** On handwritten *actas*, generic digit-restricted OCR with careful cell cleaning reaches 59.6% field accuracy (95% CI [56.6, 62.5], clustered bootstrap). This bounds what can

be expected without training and motivates fine-tuning transformer recognizers with the free labels described above.

- 5) **A head-to-head comparison with an end-to-end multimodal LLM.** Under an identical protocol, few-shot prompting of a current multimodal LLM reaches 48.4% against the pipeline’s 60.9% on the same actas, which places its value in zero-annotation bootstrapping rather than in recognition accuracy.

Code, per-field evaluation artifacts and the sampling seed are public.¹

II. RELATED WORK

Real-time detection. RT-DETR [1] showed that a carefully designed DETR outperforms comparable-scale YOLO detectors (L/X) in both accuracy and speed while removing non-maximum suppression (NMS) and its manually tuned thresholds. For tally sheets — tables of contiguous cells frequently overlapped by stamps and signatures — threshold-free, end-to-end detection is attractive, and RT-DETR’s end-to-end latency protocol is the one we will adopt to compare learned detectors (YOLOv11 vs. RT-DETR) in future work. In this paper, detection is solved by template registration, which doubles as an automatic label generator for those detectors.

Text recognition. TrOCR [4] established transformer encoder-decoder OCR with pre-trained language models, and DTrOCR [2] showed that a decoder-only generative architecture surpasses it on printed, scene and handwritten text (CER 2.38 on IAM without an external language model). These results justify transformer recognizers as the upgrade path for our handwritten-digit fields; a caveat for our setting is that language priors may “correct” digit strings toward plausible sequences, which matters when reading isolated numbers.

Unified document parsing. OmniParser [3] unifies text spotting, key information extraction and table recognition through structured-point sequences, and highlights the weakness of monolithic sequence generation: Donut [5] collapses on long tables due to error accumulation, dropping to 22.7 TEDS on PubTabNet vs. 88.83 for OmniParser. An acta is precisely a long table plus key fields; OmniParser’s explicit localization of every extracted value also matches the auditability requirement of electoral documents. We keep a modular pipeline — whose per-stage metrics are easier to diagnose — and treat unified end-to-end parsing as the alternative to compare against.

None of these works evaluates handwritten digits in Spanish on official forms with stamps and signatures; that combination is the empirical gap this project addresses.

III. DATASET

A. Source and acquisition

All documents come from the official ONPE results portal for the 2026 General Elections (first round, April 12, 2026). The portal exposes an internal API: searching for a polling

station returns its actas and their processing statuses; the acta detail lists its files, where type 1 is the scrutiny sheet; a signed S3 URL serves the PDF. Downloads use browser TLS impersonation, a conservative request rate, and a resumable manifest (`src/scrapper/` in our repository).

B. Sampling design

The universe holds $\sim 88,064$ station codes (upper bound located by exponential-plus-binary search against the API). We drew a simple random sample without replacement with a **published seed (2026)**: 100 actas, 100% hit rate, covering 23 of Peru’s department-level *ubigeo* prefixes (the Peruvian standard geographic location code) with a size-proportional distribution (Lima concentrates 36%, consistent with its electorate share). This corrects the pilot bias of sequential station codes from a single district.

C. Population heterogeneity

Classifying each PDF by page count and text layer reveals two document types: **76 handwritten actas** (single-page scans, empty text layer) and **24 born-digital STAE actas** (two-page digitally signed documents with typeset vote counts and an extractable text layer). The mix itself is a sample estimate: born-digital prevalence is 24% with a 95% Wilson interval of [16.7, 33.2] — enough to establish that the population is mixed, not enough to pin the exact share. All vision experiments run on the 76 handwritten actas; STAE actas are processed by direct text parsing (Sec. IV-E).

D. Ground truth

For every sampled station we store the officially digitized counts returned by the API: votes per political organization (38 rows), blank/null/challenged votes, totals and electorate. Form positions 04 and 14 correspond to withdrawn parties, are empty on the form and null in the API; they are excluded (a null-to-null match would spuriously inflate accuracy). This yields **43 evaluable fields per acta, 3,268 in total**.

IV. METHOD

A. Pipeline overview

(1) *Acquisition*: PDF download plus official ground truth. (2) *Preprocessing*: 300 DPI rasterization, deskew by dominant angle, CLAHE contrast normalization, non-local-means denoising (OpenCV). (3) *Region detection*: template with fiducial registration (below). (4) *Recognition*: digit-restricted OCR on cleaned cells. (5) *Structured output*: one JSON per acta with the proposal’s schema.

B. Template regions with fiducial registration

The acta is a fixed-layout form, so 46 regions (38 vote cells arranged as equally spaced rows, four summary rows, station number, electorate, citizen total, observations) are defined once as page fractions calibrated on a reference acta (Fig. 1). National sampling exposed the failure mode of fixed fractions: scanner framing varies (black margin bands, content offsets of up to ~ 184 px), shifting every cell one row off and producing

¹https://github.com/carlosperez100/onpe_actas

The image shows a sample of a Peruvian electoral acta (ACTA ELECTORAL) from the 2026 general elections. It is a form for recording votes from a specific polling station (Mesa de Sufragio N° 000001) in the Chachapoyas district. The form includes a list of political parties and candidates, with corresponding vote counts recorded in blue boxes. Red boxes highlight the header information, including the acta number, date, and location. The form also contains sections for the declaration of the acta, the declaration of the members of the mesa, and the declaration of the members of the mesa.

Fig. 1. Reference acta with the 46 template regions: 38 vote cells (blue), summary and header fields (red). Region coordinates are page fractions, applied after fiducial registration.

CER > 1. The form itself provides the fix: printed black registration squares along the border. We detect them as dark, square, high-fill connected components inside border bands; match detected marks to 15 calibrated reference positions by translation RANSAC (each detected–reference pair proposes a shift, the shift with most inliers wins), robust to missing marks and to false positives such as the printed sample digits; and estimate a partial affine transform that warps the page to the reference frame before cropping.

C. Recognition of handwritten counts

Cells are read with EasyOCR restricted to a digit vocabulary, after a cleaning cascade motivated by error analysis on a 10-acta pilot (Fig. 2): (i) an interior margin removes printed cell borders, which were previously misread as extra digits; (ii) per-cell Otsu binarization isolates pen ink and survives the alternating shaded rows of the form; (iii) the dotted intra-cell separators — as dark as ink, hence immune to thresholding — are erased by *template position* (fixed fractions of the cell width); (iv) a connected-component speckle filter removes digitization noise; (v) fields with per-digit printed

TABLE I
PILOT ABLATION ON 10 ACTAS (430 FIELDS), FIELD ACCURACY AND CER.

Configuration	Acc. (%)	CER
v1: raw digit OCR	43.95	0.524
v2: + ink isolation, borders, sub-boxes	52.09	0.419
v3: + separator erasure, Otsu, speckle	52.79	0.476
v4: + empty-cell-is-zero rule	58.37	0.420

boxes (citizen total) are read box by box. A fallback re-reads the raw crop when cleaning empties faint strokes.

D. Domain rule: empty cell means zero

Polling-station members frequently leave the cell blank when a party receives no votes. A blank cell is not an OCR failure but form semantics, so a post-processing rule maps unread party cells (and challenged votes) to zero. We report it separately to avoid crediting vision with domain knowledge.

E. Routing born-digital actas

STAE actas are parsed directly from the PDF text layer: each party name is followed by its typeset count, and summary labels precede their values. Matching extracted names to official ones by normalized string equality yields field accuracy without any image processing.

V. EXPERIMENTS

A. Metrics and statistical protocol

Field accuracy: exact match between the read integer and the official value. **CER:** character error rate between the digit strings. Fields within an acta share writer and scan quality, so the acta is the statistical cluster: confidence intervals use a **clustered bootstrap** (10,000 resamples of whole actas). All evaluation artifacts (per-field errors, per-acta metrics) ship with the repository.

B. Pilot ablation (10 actas, 430 fields)

The pilot — sequential stations of one district, hence biased but useful for component analysis — quantifies each recognition improvement (Table I).

C. National evaluation (76 handwritten actas, 3,268 fields)

Table II shows the same system on the random national sample. Fiducial registration (Sec. IV-B) is decisive: without it, framing variation severely degrades part of the sample (worst acta 14%, CER > 1). Fiducial registration adds +8.7 points and lifts the worst acta from 14% to 30%; the domain rule adds +21.6 points — nationally, blank zero-cells are far more common than in the pilot district.

Main result. Field accuracy **59.58%**, **95% CI [56.61, 62.48]** (± 2.94 points); CER **0.401**, **95% CI [0.360, 0.444]**. Per-acta accuracy averages 59.6% (SD 13.0, range 30–84%); residual variance stems from handwriting and digitization quality, no longer from alignment. The pilot figure (58.4%) happens to be close, but only the national estimate has a valid margin of error and offers national coverage.

TABLE II
NATIONAL RANDOM SAMPLE, 76 HANDWRITTEN ACTAS (3,268 FIELDS).

Configuration	Acc. (%)	CER
Fixed template, raw OCR	26.81	0.724
Fixed template + zero rule	50.86	0.482
Registered (v5), raw OCR	38.00	0.617
Registered (v5) + zero rule	59.58	0.401

TABLE III
MODULAR PIPELINE VS. END-TO-END LLM FEW-SHOT ON THE SAME 15 HANDWRITTEN ACTAS (645 FIELDS).

Method	Acc. (%)	CER
Pipeline (registered template + digit OCR)	60.93	0.395
LLM few-shot, end-to-end (2 exemplars)	48.37	0.591

Born-digital actas. Text-layer parsing of the 24 STAE actas reaches **95.12%** (955/1,004 fields); the remaining errors stem from name matching, not from reading. Twenty-four actas suffice here because parsing is deterministic: between-acta variance is below binomial ($ICC \approx 0$, design effect 1.0), so the clustered bootstrap interval is [94.5, 95.7], nine times tighter than the handwritten estimate, whose design effect is 3.07 for a comparable field count. The 35.5-point gap (95% CI [32.6, 38.5]) quantifies the value of routing by document type.

D. End-to-end multimodal LLM baseline

A natural alternative to the modular pipeline is to hand the whole page to a multimodal LLM together with a few solved examples, letting it localize and read in a single step. We evaluate that baseline²: the model receives the full acta image plus two image–JSON exemplars and must return the vote-count JSON, with no template, no registration and no OCR engine. Both systems are scored on the same 15 handwritten actas (645 fields) with the same protocol (Table III).

The pipeline wins by 12.6 points. The LLM parses the form structure competently, but it is not trained for handwritten Peruvian digits, and it degrades faster on poor scans: on the two worst actas it drops to 20.9% and 23.3% where the pipeline still reads 34.9% and 53.5%, since cropping gives the recognizer an isolated cell while the LLM must fight the surrounding noise. This is consistent with DTrOCR [2]: strong handwriting recognition comes from task-specific training, not from general capability. Its advantage is orthogonal to accuracy — it required zero detection engineering, whereas the pipeline needed template calibration and fiducial registration — which makes it useful as a bootstrap annotator for a new form layout, and as a pre-labeler whose output a human corrects, rather than as a replacement for a specialized recognizer.

E. Error analysis

Remaining errors concentrate in: isolated thin digits missed by the detector stage of the OCR engine; strokes invading the

erased separator bands; shape confusions ($8 \leftrightarrow 18$, $7 \leftrightarrow 9$) on wide handwriting; and multi-box fields. These are recognition errors, not localization errors — exactly the component that fine-tuning replaces.

VI. CONCLUSIONS AND FUTURE WORK

A classical pipeline over real, nationally sampled Peruvian actas achieves 59.6% field accuracy (95% CI [56.6, 62.5]) on handwritten documents and 95.1% on born-digital ones, with an evaluation protocol that needs no manual annotation. Three design lessons stand out: register templates on the form’s fiducial marks before any fixed-coordinate cropping; separate domain semantics (blank = 0) from vision performance; and classify the document type before choosing a processing path. A fourth lesson comes from the LLM baseline: prompting a general multimodal model end-to-end is cheaper to build but 12.6 points less accurate (48.4%), so on this task generality does not yet substitute for specialization. The quantified ceiling of generic OCR motivates the next steps: fine-tune a transformer recognizer [2], [4] on the free field–value pairs (~ 43 per acta, $\sim 43,000$ from 1,000 actas), train learned detectors (YOLOv11, RT-DETR [1]) on labels auto-generated by the registered template, compare the modular pipeline against unified parsers [3], [5], and use the LLM as a pre-labeler in the human-correction loop that produces the fine-tuning set.

REFERENCES

- [1] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, “DETRs beat YOLOs on real-time object detection,” in *Proc. IEEE/CVF CVPR*, 2024, pp. 16965–16974.
- [2] M. Fujitake, “DTrOCR: Decoder-only transformer for optical character recognition,” in *Proc. IEEE/CVF WACV*, 2024, pp. 8025–8035.
- [3] J. Wan, S. Song, W. Yu, Y. Liu, W. Cheng, F. Huang, X. Bai, C. Yao, and Z. Yang, “OmniParser: A unified framework for text spotting, key information extraction and table recognition,” in *Proc. IEEE/CVF CVPR*, 2024, pp. 15641–15653.
- [4] M. Li, T. Lv, J. Chen, L. Cui, Y. Lu, D. Florencio, C. Zhang, Z. Li, and F. Wei, “TrOCR: Transformer-based optical character recognition with pre-trained models,” in *Proc. AAAI*, 2023, pp. 13094–13102.
- [5] G. Kim, T. Hong, M. Yim, J. Nam, J. Park, J. Yim, W. Hwang, S. Yun, D. Han, and S. Park, “OCR-free document understanding transformer,” in *Proc. ECCV*, 2022, pp. 498–517.

²Gemini 2.5 Flash, free tier, accessed July 2026. Code: `src/recognition/llm_ocr.py`.



Fig. 2. Ink isolation on problem cells (top: crop; bottom: cleaned). Dotted separators and printed borders — previously read as digits (e.g. 18 → 418) — are removed by threshold, template-positioned erasure and speckle filtering.